

## ITSS SOFTWARE DEVELOPMENT

### 0. INTRODUCTION TO COURSE

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### Course objectives

- **Design** effective program structures with
  - appropriate modularity
  - separation of abstraction and implementation concerns
  - use of standard design patterns
- Use modern **programming** languages and **unit test** framework effectively
  - type systems, objects and classes, modularity
  - identity and equality, exceptions and assertions
- Gain experience with contemporary software tools
  - design tools
  - integrated development environments (IDE)
  - test frameworks, version control
  - documentation processing tools

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### Main topics

- Software Development Process
- Object-Oriented Analysis and Design
- Design Concepts: Coupling and Cohesion
- Design Principles, Design Patterns
- Software Quality, Software Testing
- Techniques for Effective Programming

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### Programming language/tools

- Software design tool: Astah
  - Free for students
- Programming language: Java
- IDE: Eclipse
- Version control: Github
- Test framework: JUnit
- Architectural model / pattern: 3 tiers / MVC



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## Assessment

- Mid-term score: 50%
  - Case Study (Hands-on Lab)
    - Work in individual
  - Capstone Project
    - Work in groups, but individual score
  - *Reward or Punish*
    - *Quizzes/Exercises/Attendance at class:  $\pm 1$*
- Final score: 50%
  - Written exam

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## Reference books

- D. Budgen. **Software Design**, 2<sup>nd</sup> Edition. Addison-Wesley. 2004.
- Cay Horstmann. **Object-Oriented Design and Patterns**. John Wiley & Sons, Inc. 2006
- Joshua Bloch. **Effective Java**, 2<sup>nd</sup> ed. Addison-Wesley, 2008
- Boris Beizer. **Software Testing Techniques**, 2nd Edition. International Thomson Computer Press
- Erich Gamma, Richard Helm, Ralph Johnson, John Vlissides. **Design Patterns: Elements of Reusable Object-Oriented Software**. 1994. Addison-Wesley Professional.
- Robert Martin. **Agile Software Development: Principles, Patterns, and Practices**. 2003. Pearson Education.
- Robert Martin. **Clean Code: A Handbook for Agile Software Craftsmanship**. 2008. Prentice Hall.

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## Course Materials

- Lecture notes for students (pdf): Slides in 4-page handouts
- Assignments, Project descriptions
- Interaction channels:
  - MS Teams
  - <https://github.com>
    - Add to your project member: trangntt.for.student@gmail.com

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## Naming convention

- Naming your private individual repo (Lab)
  - MTeamName-StudentId.StudentName
  - E.g. ISD.ICT.20221-20191234.NguyenVanHa  
or ISD.VN.20221-20199283.VuHuongLan
- Naming your private group repo (Capstone Project)
  - **TeamName-GroupNo**
  - E.g ISD.ICT.20221-05 or ISD.VN.20221-10
- Monitor?

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## Introduce yourselves

- Full name
- Experience in Computer Science
  - Operating System
  - Programming Languages
  - (Mini-)Projects
  - ...
- Strength / Weakness
- A course you like best / hate
- Desire to study in this course

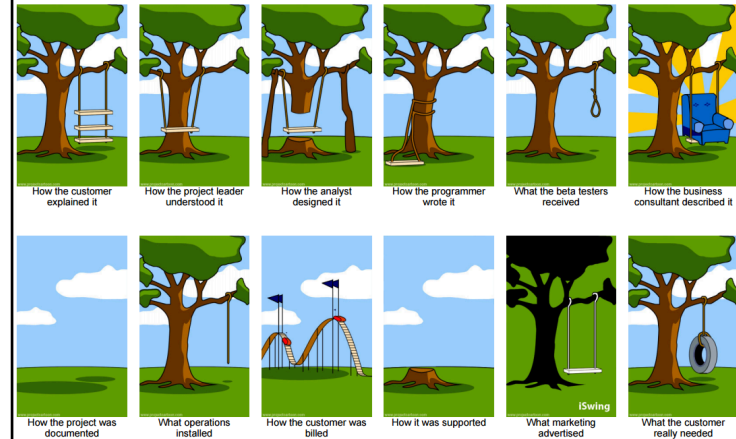


About Me

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## What is the real software to be built?



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